RetailPulse

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AI-Powered Customer Analytics & Demand Forecasting Platform

End-to-End Data Science & Analytics Solution

for Retail Demand Prediction & Customer Insights

Predictive Demand · Customer Segmentation · Churn Analysis · Inventory Optimization

"Retailers lose billions due to poor demand forecasting and stock mismanagement. RetailPulse uses advanced ML to predict demand, segment customers, detect churn, and optimize inventory."

30–50%	15–25%	MAPE ≤ 12%	10M+
Stockout Reduction	Revenue Increase	Forecast Accuracy	Transactions/Mo

Advanced Data Science & Analytics Project · Zidio Development · Portfolio / Interview Reference

Author · Date · Version · Domain

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1 Business Case & Production Objectives

Mission

RetailPulse is an end-to-end data science platform that ingests sales, customer, and inventory data to deliver accurate demand forecasts, customer segmentation, churn prediction, and inventory optimization recommendations for retail businesses. The platform addresses the multi-billion dollar problem of retail stockouts and overstock by applying advanced machine learning and analytics.

Why RetailPulse Matters

Retail clients need data-driven decisions to reduce waste and maximize profit. RetailPulse provides a complete analytics solution that Zidio Development can offer to supermarket chains, fashion retailers, and e-commerce companies. The platform bridges the gap between raw transactional data and actionable business intelligence.

30–50%	15–25%	≤ 12%	<5 min	10M+
Stockout Reduction	Revenue Increase	Model Accuracy (MAPE Target)	Daily Batch Processing	Monthly Transactions

Strategic Business Objectives

Demand Forecasting Accuracy	Achieve MAPE ≤ 12% for 30-day ahead predictions using a Prophet + LSTM ensemble, enabling confident purchasing decisions.
Customer Retention	Identify at-risk customers with AUC-ROC ≥ 0.88, enabling proactive retention campaigns that improve lifetime value.
Inventory Efficiency	Reduce both overstock and understock scenarios by 25–40% through ML-driven reorder recommendations.
Scalable Processing	Process 10M+ transactions per month with daily batch jobs completing in under 5 minutes using optimized ETL pipelines.
Full Observability	Implement complete ML flow tracking, model versioning, and drift detection for production reliability.

2 Core Functional Requirements

The following six core capabilities define the full scope of RetailPulse. Each requirement includes detailed acceptance criteria and quantified production metrics.

ID	Capability	Description & Business Value	Acceptance Criteria
F-01	**Data Ingestion & Cleaning**	Ingest sales, customer, and inventory data from multiple sources with automated ETL and data quality checks	Automated pipeline; data quality validation with Great Expectations; <5 min batch run
F-02	**Customer Segmentation**	RFM + behavioral segmentation using K-Means / DBSCAN clustering with business interpretation	6–8 meaningful segments; silhouette score > 0.45; labeled with business personas
F-03	**Demand Forecasting**	Time-series forecasting with Prophet + LSTM ensemble model for 30-day ahead predictions	MAPE ≤ 12%; 30-day ahead predictions; confidence intervals provided
F-04	**Churn Prediction**	XGBoost classification model with SHAP explainability to identify at-risk customers	AUC-ROC ≥ 0.88; Precision@top-20% ≥ 0.75; SHAP feature plots
F-05	**Inventory Optimization**	Recommend reorder quantities using forecasted demand and safety stock calculations	Reduce overstock/understock by 25–40%; daily reorder point updates
F-06	**Interactive Analytics Dashboard**	Streamlit multi-page dashboard with visualizations, what-if analysis, and export functionality	Real-time insights; exportable CSV/PDF reports; load time <3s

3 Production Technology Stack – 2026

RetailPulse is built on a modern, production-grade Python ecosystem with containerized deployment, full MLOps observability, and cloud-native scalability.

Layer	Primary Technology	Rationale & Alternatives
Language	**Python 3.11**	Industry-standard data science ecosystem; type hints, performance improvements
Data Processing	**Pandas, NumPy, Scikit-learn**	Core data manipulation, vectorized operations, and ML primitives
Forecasting	**Prophet + LSTM (PyTorch)**	Hybrid time-series: Prophet handles seasonality, LSTM captures non-linear patterns
Dashboard	**Streamlit**	Rapid interactive analytics UI; alt: Dash, Gradio, FastAPI + React
Experiment Tracking	**MLflow**	Model versioning, reproducibility, artifact storage; alt: Weights & Biases
Database	**PostgreSQL + Redis**	Structured transactional data + high-speed caching for dashboard queries
Containerization	**Docker**	Consistent multi-stage builds; environment reproducibility
Orchestration	**Kubernetes**	Scalable production deployment; horizontal pod autoscaling
Monitoring	**Prometheus + Grafana + Evidently AI**	Drift detection, performance monitoring, alerting dashboards
Pipeline Orchestration	**Apache Airflow**	Automated daily retraining and ETL scheduling
CI/CD	**GitHub Actions**	Automated testing, linting, and deployment on merge
Cloud	**AWS / GCP**	EC2/GKE for compute; S3/GCS for artifact storage; RDS for PostgreSQL

4 System Architecture Overview

RetailPulse follows a layered MLOps architecture — from raw data ingestion through feature engineering, model training, serving, and monitoring. The diagram below illustrates the end-to-end data and ML pipeline.

DATA SOURCES	POS Systems · E-commerce API · Inventory DB · CRM Data · External (Weather/Holidays)
INGESTION & ETL	Apache Airflow DAGs · Great Expectations Validation · PostgreSQL Storage · Redis Caching
FEATURE ENGINEERING	RFM Scoring · Rolling Statistics · Time-Series Decomposition · Lag Features · Behavioral Signals
ML MODELS	Prophet + LSTM Forecasting · K-Means / DBSCAN Segmentation · XGBoost Churn Classifier · Inventory Optimizer
EXPERIMENT TRACKING	MLflow Model Registry · Optuna Hyperparameter Tuning · SHAP Explainability · Evidently AI Drift Detection
SERVING & DASHBOARD	Streamlit Multi-Page App · What-If Analysis · CSV / PDF Export · Real-Time Alerts
MONITORING & OPS	Prometheus Metrics · Grafana Dashboards · Automated Retraining · GitHub Actions CI/CD

Data flows from ingestion through ETL → feature store → model training (tracked in MLflow) → serving via Streamlit → monitored by Prometheus/Grafana with automatic drift-triggered retraining via Airflow.

5 28-Day Execution Plan

The project is structured as four one-week sprints, each building on the previous. Each week ends with a concrete checkpoint deliverable.

Week 1	Data Exploration & Preparation
Day 1	Dataset selection; Initial EDA notebook: distributions, missing values, correlation heatmaps
Day 2	Data cleaning, feature engineering (RFM scores, rolling stats); Great Expectations validation
Day 3	Customer segmentation: K-Means and DBSCAN; cluster evaluation and business labeling
Day 4	Time-series data prep for forecasting; stationarity tests (ADF/KPSS); decomposition
Day 5	Baseline Prophet model training and initial evaluation
Day 6	LSTM implementation with PyTorch Lightning; training loop setup
Day 7	**CHECKPOINT: EDA report, cleaned dataset, baseline models logged in MLflow**

Week 2	Advanced Modeling & Churn Prediction
Day 8	Hybrid forecasting model: Prophet + LSTM weighted ensemble
Day 9	Churn prediction with XGBoost; SHAP explainability plots; feature importance
Day 10	Inventory optimization logic using forecasted demand + safety stock formulas
Day 11	Hyperparameter tuning with Optuna; cross-validation and model selection
Day 12	Drift detection setup using Evidently AI; baseline data profiling
Day 13	Automated retraining pipeline with Apache Airflow DAGs
Day 14	**CHECKPOINT: Forecasting and churn models ready; optimization logic implemented**

Week 3	Dashboard & Analytics Layer
Day 15	Streamlit dashboard skeleton: multi-page layout, navigation, theming
Day 16	Demand forecasting visualizations; what-if analysis (price/promo sliders)
Day 17	Customer segmentation view; churn risk scores and retention recommendations
Day 18	Inventory optimization UI: reorder recommendations with confidence ranges
Day 19	Real-time metrics stream; threshold-based alert notifications
Day 20	Export functionality: CSV data exports and styled PDF report generation
Day 21	**CHECKPOINT: Fully interactive dashboard with all business insights accessible**

Week 4	Deployment & Production Polish

Day 22	Docker multi-stage builds for application services
Day 23	Kubernetes manifests, Helm charts, and HPA configuration
Day 24	GitHub Actions CI/CD pipeline: lint → test → build → deploy
Day 25	Cloud deployment on AWS (EC2 + RDS + S3) or GCP (GKE + Cloud SQL)
Day 26	Monitoring setup: Prometheus scrapers, Grafana dashboard JSON
Day 27	Load testing (Locust); final accuracy validation against holdout set
Day 28	Final QA, README polishing, demo video recording, PDF export, submission

6 Technical Highlights & Model Details

Demand Forecasting: Hybrid Prophet + LSTM Ensemble

The forecasting engine combines Facebook Prophet (capturing trend, seasonality, and holiday effects) with a PyTorch Lightning LSTM network (learning non-linear temporal patterns and residuals). Final predictions are a weighted ensemble of both models, with weights optimized via Optuna on validation data.

Model Component	MAE	RMSE	MAPE	Notes
Prophet (baseline)	–	–	~16%	Strong on seasonality & trends
LSTM (residual learner)	–	–	~14%	Captures non-linear patterns
Ensemble (weighted avg)	–	–	≤ 12%	Target: production threshold

Customer Segmentation: RFM + Behavioral Clustering

Customers are scored on Recency (days since last purchase), Frequency (purchase count), and Monetary value (total spend). These RFM scores are combined with behavioral features (category affinity, channel preference, return rate) before applying K-Means and DBSCAN to produce 6–8 actionable segments.

Segment	Profile	Strategy
Champions	High R, F, M — recent, frequent, high-spend	Reward & upsell; brand ambassadors
Loyal Customers	High F & M, moderate R	Loyalty program; early access offers
Potential Loyalists	Recent buyers with moderate frequency	Nurture with targeted campaigns
At-Risk Customers	High historical value, low recency	Win-back emails; special discounts
Churned	Low R, F, M — no recent activity	Last-chance reactivation or sunset
New Customers	Very recent, low frequency	Onboarding sequence; first repeat purchase push

Churn Prediction: XGBoost + SHAP

An XGBoost binary classifier predicts 90-day churn probability. SHAP (SHapley Additive exPlanations) values explain each prediction at the individual customer level, providing interpretable outputs for marketing teams. Key features include days since last purchase, purchase frequency trend, average order value change, and support ticket history.

≥ 0.88	≥ 0.75	≥ 0.80	≥ 0.78
AUC-ROC	Precision@20%	F1 Score	Recall

7 MLOps & Production Readiness

Production ML systems require far more than model accuracy. RetailPulse implements a full MLOps maturity level 2 pipeline with automated retraining, drift detection, and comprehensive observability.

MLflow Experiment Tracking	All experiments, parameters, metrics, and artifacts are tracked in MLflow. Model registry manages staging → production promotion with version control and rollback capabilities.
Evidently AI Drift Detection	Automated data drift reports compare incoming feature distributions against training baselines. Alerts trigger when PSI > 0.2 on critical features, initiating the retraining pipeline.
Airflow Automated Retraining	Daily DAG validates data quality, retrains models when drift is detected, evaluates on holdout set, and auto-promotes to production if MAPE improves by >1%.
Optuna Hyperparameter Optimization	Bayesian optimization with 50+ trials per model training run. Best hyperparameters stored in MLflow for reproducibility. Pruning enabled to terminate underperforming trials early.
SHAP Model Explainability	SHAP summary plots, waterfall charts, and force plots generated for churn and forecasting models. Business teams can understand model decisions at both global and individual levels.

8 Deployment & Operations

Component	Technology	Configuration
Containerization	Docker (multi-stage)	Separate build/runtime stages; <200MB final image
Orchestration	Kubernetes	HPA enabled; min 2 / max 10 replicas based on CPU
CI/CD Pipeline	GitHub Actions	On PR: lint (ruff) → test (pytest) → build → push ECR
Cloud Infrastructure	AWS EC2 + RDS + S3	EC2 t3.xlarge; RDS PostgreSQL 15; S3 for model artifacts
DNS & SSL	Route 53 + ACM	HTTPS enforced; auto-renewing TLS certificates
Monitoring	Prometheus + Grafana	Custom dashboards; PagerDuty alerts on SLA breach
Log Management	CloudWatch Logs	Structured JSON logs; 30-day retention; error alerting

9 Security & Privacy Highlights

Data Anonymization	Customer PII anonymized at ingestion; reversible tokenization for analytics; GDPR-compliant data retention policies.
Role-Based Access Control	Streamlit dashboard enforces RBAC: Analyst / Manager / Admin roles with granular feature access per tier.
Secure API Endpoints	All API routes protected with JWT authentication; token expiry 1 hour; refresh token rotation.
Audit Logging	Full audit trail for sensitive operations (data export, model promotion, user access) stored in immutable logs.
Secrets Management	No secrets in code or Git; AWS Secrets Manager for DB credentials; environment variables for runtime config.

10 Challenges & Industry Best Practices

Building a production-grade retail analytics platform involves navigating several real-world data science challenges. The following summarizes key obstacles encountered and the solutions applied.

Non-Stationary Time Series	Challenge: Retail sales data exhibits trend shifts, promotional spikes, and structural breaks that violate stationarity assumptions. Solution: ADF and KPSS tests to diagnose stationarity; STL decomposition to isolate trend/seasonal components; differencing and log-transform as preprocessing; Prophet's changepoint detection to handle structural breaks adaptively.
Accuracy vs. Interpretability	Challenge: Complex ensemble models (LSTM + Prophet) offer high accuracy but are black-box to business stakeholders who need to trust and act on outputs. Solution: SHAP values for churn model; Prophet's built-in component plots for forecasting; Streamlit dashboard surfaces human-readable explanations alongside predictions.
Class Imbalance in Churn	Challenge: Churn events are rare (~5–15% of customers), causing naive models to predict no-churn for everyone and achieve misleadingly high accuracy. Solution: SMOTE oversampling; class_weight='balanced' in XGBoost; evaluation on AUC-ROC and Precision@top-20% rather than raw accuracy; threshold tuning on validation set.
Data Drift in Production	Challenge: Consumer behavior shifts (seasonal patterns, economic changes) cause model performance to degrade over time without detection. Solution: Evidently AI monitors PSI (Population Stability Index) and feature drift weekly; Airflow triggers automated retraining when PSI > 0.2; MLflow tracks model version history for rollback.
Scalable Pipeline Performance	Challenge: Processing 10M+ transactions monthly while maintaining <5-minute daily batch jobs. Solution: Vectorized Pandas operations; Redis caching for dashboard queries; partitioned PostgreSQL tables by date; Kubernetes horizontal pod autoscaling; pre-aggregated feature tables.

11 Personal Reflection & Future Roadmap

Key Learnings

1. Production ML is 20% modeling and 80% data engineering, MLOps, and infrastructure — RetailPulse reinforced the importance of robust ETL pipelines and monitoring as the true foundation of reliable AI systems.

2. Interpretability is not optional in business contexts. SHAP values transformed the churn model from a black box into a tool that marketing teams could act on with confidence.

3. Ensemble methods combining statistical (Prophet) and deep learning (LSTM) approaches consistently outperform single-model solutions for retail time-series with mixed trend and non-linear patterns.

4. Drift detection and automated retraining are essential in retail, where consumer behavior changes seasonally, economically, and in response to external events.

5. Streamlit enabled rapid iteration from prototype to polished dashboard — demonstrating that data scientists can build production-quality analytics interfaces without a dedicated frontend team.

Industry Best Practices Applied

Practice	Application in RetailPulse
MLflow for Reproducibility	Every experiment, parameter, metric, and model artifact logged. Complete audit trail from raw data to deployed model.
Evidently AI for Drift Detection	Automated data and model drift reports with configurable alert thresholds and retraining triggers.
Airflow for Orchestration	Robust DAG-based scheduling with dependency management, retries, and alerting for pipeline failures.
Great Expectations for Data Quality	Automated data validation with documented expectations; failures halt the pipeline before bad data corrupts models.
SHAP for Explainability	Model decisions are transparent and auditable — critical for retail clients who need to justify AI recommendations.

Future Roadmap

v2.1 – Real-Time Streaming	Replace batch ETL with Apache Kafka + Spark Streaming for sub-minute demand signal updates.
v2.2 – LLM-Powered Insights	Integrate an LLM (Claude API) to generate natural language summaries of anomalies and recommendations directly in the dashboard.
v2.3 – Multi-Tenant SaaS	Convert to a multi-tenant SaaS platform with isolated data environments and self-service onboarding for retail clients.

v2.4 – Computer Vision Integration	Add shelf-image analysis for real-time stock visibility using YOLOv8 — connecting physical inventory signals to the forecasting engine.
v3.0 – Causal ML	Replace correlation-based churn signals with causal inference models (DoWhy) to separate genuine churn drivers from confounders.

RetailPulse — AI-Powered Customer Analytics & Demand Forecasting Platform Prepared for Zidio Development · Data Science & Analytics Domain · March 2026 · Version 2.0

Crafted with precision and modern data science principles · Advanced Industry-Grade Project